Ideation Phase

Brainstorm & Idea Prioritization Template

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template

Brainstorming provided a free and open environment that encouraged every member of the team to participate in the creative-thinking process leading to problem solving for the project, **Electricity Consumption Analysis**. Prioritizing volume over value, innovative ideas were welcomed and discussed, allowing the team to identify multiple approaches for analyzing electricity usage data. Every member actively contributed suggestions related to data collection, visualization, dashboard development, database management, and web application integration.

This document explains how the team followed a structured brainstorming and idea prioritization process to move from an initial project idea to a well-defined and practical solution. Through collaboration and discussion, the team selected the most suitable ideas that could be successfully implemented using Excel, MySQL, Tableau, Python Flask, and GitHub.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The project team assembled to discuss various real-world problems that could be solved using data analytics and visualization. Several project ideas such as water consumption analysis, traffic analysis, weather forecasting, and electricity consumption analysis were discussed. After comparing their feasibility, availability of datasets, implementation complexity, and real-world usefulness, the team unanimously selected **Electricity Consumption Analysis**.

Relevant information about electricity datasets, Tableau dashboard examples, SQL database implementation, and web application development was shared among all members. Each member contributed valuable suggestions regarding data preprocessing, dashboard creation, interactive visualizations, and business insights before finalizing the project scope.

PROBLEM STATEMENT

How might we analyze electricity consumption across different states, regions, and time periods using data analytics, so that electricity boards, government agencies, energy planners, and decision-makers can understand consumption patterns, identify high-demand areas, monitor regional electricity usage, and make informed decisions for efficient energy planning and sustainable resource management?

Key Rules of Brainstorming Followed

✓ Stay on Topic All discussions and ideas focused on electricity consumption analysis, energy usage patterns, regional comparison, and interactive visualization.
✓ Encourage Wild Ideas Creative ideas such as predictive analytics, smart dashboards, GIS-based maps, forecasting models, and web-based reporting were welcomed before evaluation.
✓ Defer Judgment No idea was rejected during the initial brainstorming session. Every suggestion was recorded first and evaluated later based on feasibility.
✓ Listen to Others Each team member carefully listened to ideas shared by others, discussed improvements, and collaboratively refined the project plan.
✓ Go for Volume The objective was to generate as many useful ideas as possible before selecting the final implementation approach.
✓ Be Visual Ideas were grouped into meaningful categories using diagrams, tables, and workflow planning to improve understanding and decision-making.

Step-2: Brainstorm, Idea Listing and Grouping

Individual Idea Listing

Each team member independently suggested ideas related to the project before discussing them as a group.

Team Member 1	**Team Member 2**	**Team Member 3**	**Team Member 4**
Collect electricity consumption dataset.	Compare electricity usage between regions.	Design Tableau dashboards.	Develop Flask web application.
Clean missing and duplicate records.	Identify highest electricity-consuming states.	Create interactive filters.	Publish project on GitHub.
Store data in MySQL database.	Analyze monthly consumption trends.	Build Tableau Story.	Deploy Tableau Public dashboard.
Create KPIs and summary cards.	Compare state-wise electricity usage.	Design geographic maps.	Prepare complete documentation.
Build SQL queries for analysis.	Study yearly consumption patterns.	Create Top 10 consumer visualization.	Create project presentation and demo.

Grouped Ideas (Thematic Clusters)

Cluster A – Data Collection & Preparation • Collect electricity consumption dataset. • Clean duplicate and missing values. • Import dataset into MySQL. • Verify data quality. • Prepare dataset for visualization.
Cluster B – Data Analysis • Analyze state-wise electricity consumption. • Compare electricity usage across regions. • Identify highest electricity-consuming states. • Analyze monthly electricity trends. • Generate statistical summaries.
Cluster C – Dashboard & Visualization • Create interactive Tableau dashboard. • Develop geographic maps. • Create KPI indicators. • Design Tableau Story. • Develop business insights.
Cluster D – Deployment & Documentation • Publish dashboard on Tableau Public. • Build Flask web application. • Upload source code to GitHub. • Prepare project documentation. • Record project demonstration video.

Step-3: Idea Prioritization

The team evaluated all collected ideas based on their importance, feasibility, available dataset, implementation time, and project objectives. The ideas were categorized into four groups.

Plan Next	Do First
• Electricity Consumption Forecasting • AI-based Demand Prediction • Real-time Data Integration	• Data Cleaning & Preparation • MySQL Database Creation • Tableau Dashboard Development • State-wise Electricity Analysis • Monthly Trend Analysis
Reconsider	**Fill-in Tasks**
• Machine Learning Prediction Models • IoT Smart Meter Integration • Live Electricity Monitoring Dashboard	• Tableau Story Creation • Flask Web Application • GitHub Repository • Tableau Public Deployment • Project Documentation

The ideas placed under the **Do First** category became the foundation of the project. These activities included data cleaning, SQL database creation, Tableau dashboard development, business insight generation, and Tableau Story creation. Supporting activities such as Flask web application development, GitHub documentation, and Tableau Public deployment were completed alongside the main implementation. Advanced features such as machine learning forecasting and real-time electricity monitoring were identified as future enhancements due to time limitations and dataset availability.